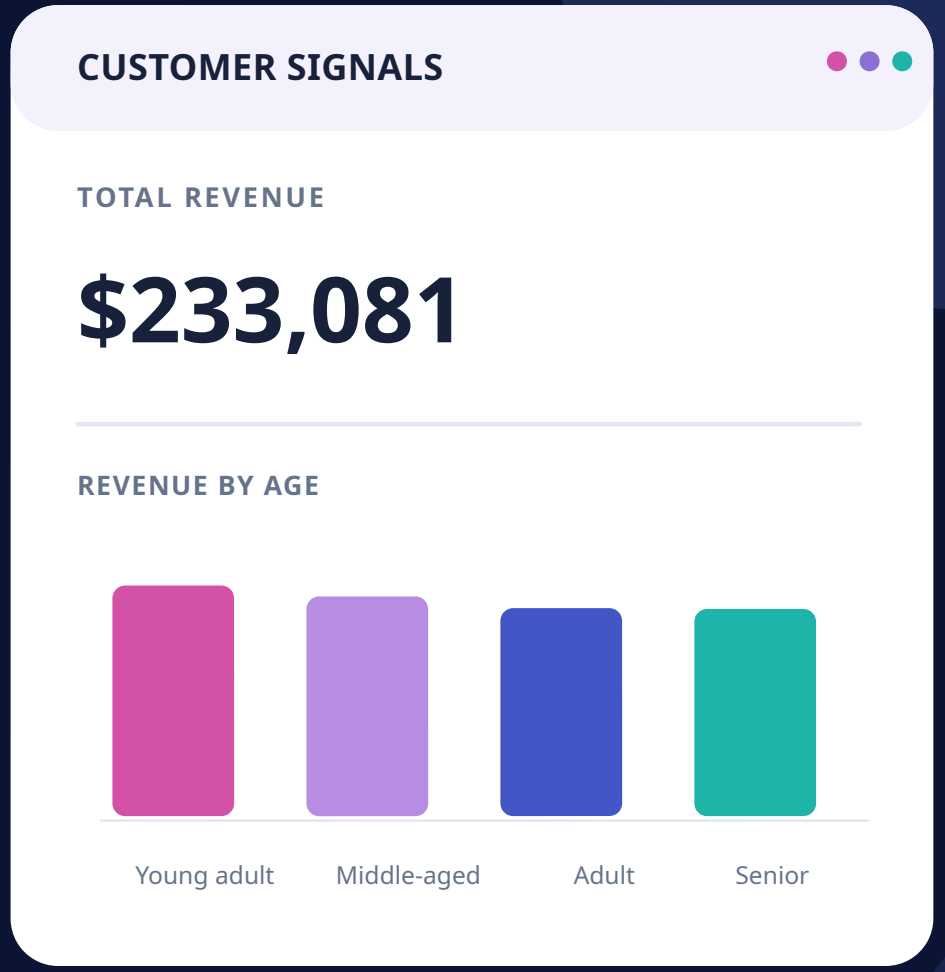
DATA ANALYTICS / PORTFOLIO CASE STUDY

CUSTOMER SHOPPING BEHAVIOR

Turning 3,900 purchase records into actionable customer insights

PYTHON | POSTGRESQL | POWER BI



TRANSACTIONAL ANALYSIS • CUSTOMER STRATEGY

Four numbers tell the opening story

A descriptive view of purchase volume, revenue, order value, and subscription participation.

3,900

Purchase records

18 attributes analyzed

$233,081

Recorded revenue

Across all transactions

$59.76

Average purchase

USD per purchase

27.0%

Subscribed

1,053 of 3,900 records

Revenue is not evenly split by gender

Male-tagged purchases account for 67.7% of recorded revenue. This does not measure spend per person.

A sizable subscription audience remains

73.0% of records are non-subscribed, while subscriber and non-subscriber average order values are similar.

Business question: How can we convert demand into more durable, profitable customer relationships?

What we analyzed

Transactional attributes span customer profiles, purchase context, and shopping habits.

DATASET SIZE

3,900

purchase records

18 data columns

37 missing review ratings

01

CUSTOMER PROFILE

Age • Gender • Location • Subscription

02

PURCHASE DETAILS

Item • Category • Amount • Season • Size • Color

03

SHOPPING BEHAVIOR

Discount • Frequency • Prior purchases • Rating • Shipping

A reproducible Python → SQL → Power BI workflow

Cleaning and feature engineering happen before structured business queries and visualization.

01

Load & inspect

pandas, df.info(), summary statistics

02

Clean the data

Category-median rating imputation; snake_case

03

Engineer fields

Age groups and purchase-frequency days

04

Analyze in SQL

PostgreSQL queries, segments and rankings

05

Visualize

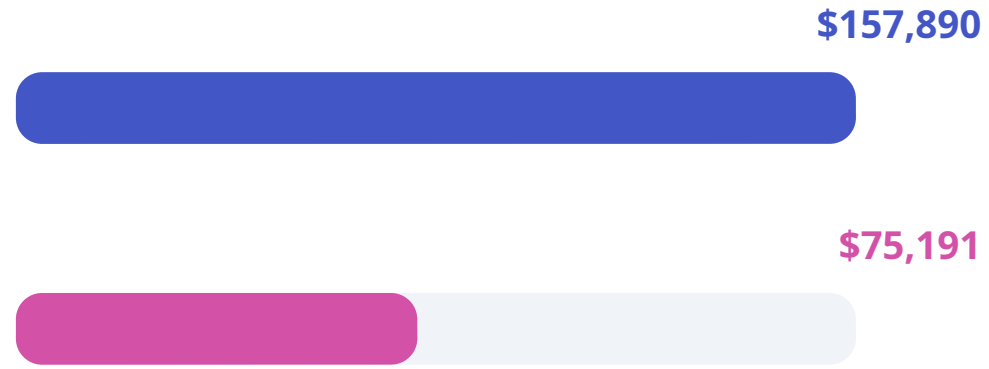
Power BI dashboard and recommendations

Consistency check: discount_applied and promo_code_used were redundant; promo_code_used was removed.

Where does the revenue come from?

Revenue by recorded gender label; totals describe transactions, not per-customer value.

REVENUE BY GENDER



Gender labels are attributes in the source dataset; group sizes are not shown here.

SHARE OF TOTAL REVENUE

67.7%

Male-tagged transactions



32.3% female

Subscriptions have headroom to grow

The dataset contains far more non-subscribed than subscribed purchase records.

SUBSCRIPTION MIX

73.0%

Not subscribed



No · 2,847 records

Yes · 1,053 records

AVERAGE PURCHASE AMOUNT

$59.87

Non-subscribers

$59.49

Subscribers

Only $0.38 separates the two averages.

“Loyal” dominates the purchase-history segments

Customers were classified as New, Returning, or Loyal based on their prior-purchase history.

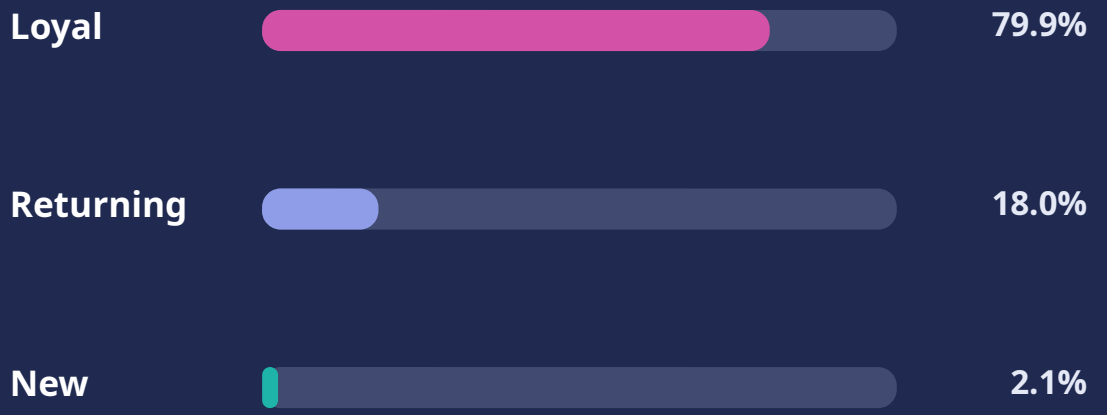
79.9%

Loyal segment

3,116 of 3,900 records

Segmentation uses a historical purchase field; it does not establish retention over time.

SEGMENT DISTRIBUTION

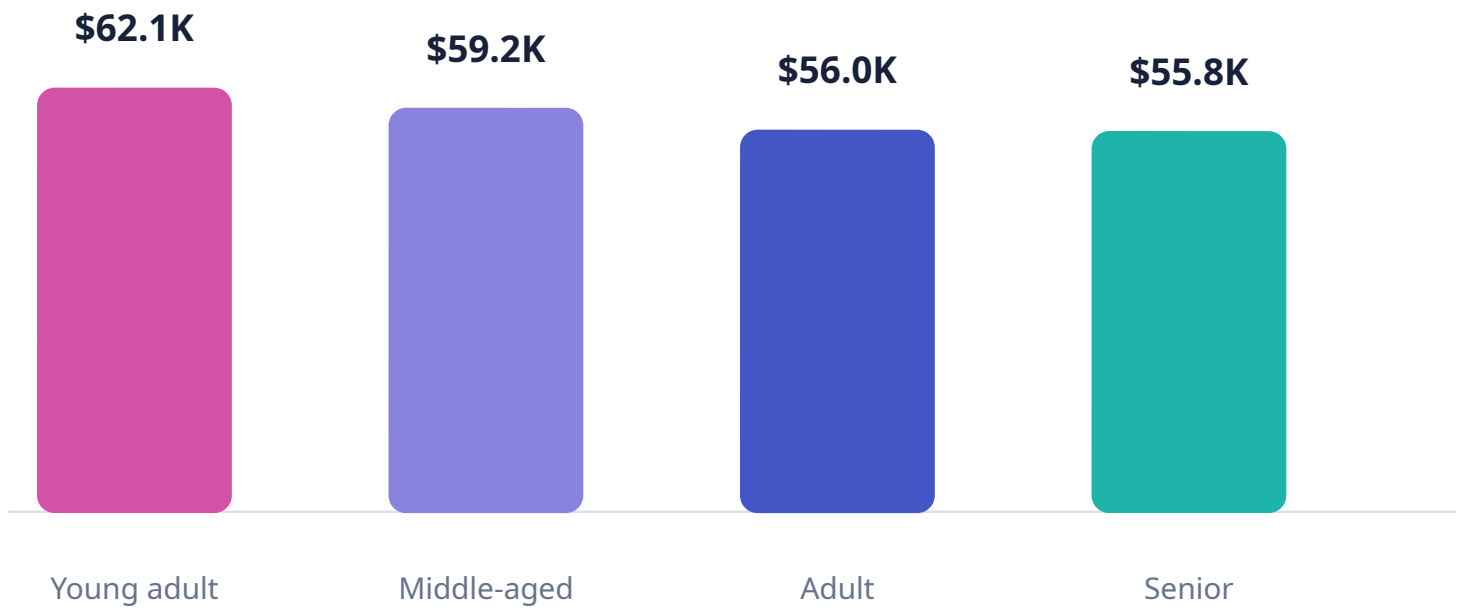


Among records with >5 prior purchases, 958 are subscribed and 2,518 are not (27.6% subscribed).

Revenue is spread across age segments

The top age band leads, but the four groups contribute relatively similar revenue totals.

REVENUE BY AGE GROUP



TOP GROUP

26.7%

of total revenue

Only $6.4K separates the highest and lowest revenue age groups.

Best sellers differ by category

Top product counts are ranked within each merchandise category.

ACCESSORIES

Jewelry

171

purchases

ALSO HIGH-RANKING

Sunglasses	161
Belt	161

CLOTHING

Blouse

171

purchases

ALSO HIGH-RANKING

Pants	171
Shirt	169

FOOTWEAR

Sandals

160

purchases

ALSO HIGH-RANKING

Shoes	150
Sneakers	145

OUTERWEAR

Jacket

163

purchases

ALSO HIGH-RANKING

Coat	161

High ratings and discount reliance are different signals

Product satisfaction and the use of promotions should be managed separately.

HIGHEST AVERAGE REVIEW RATINGS



Average review rating, scale 1–5

HIGHEST DISCOUNT USAGE RATES



Discount usage rate among that product's purchases

Express shipping is linked to a slightly larger basket

A descriptive comparison of the average purchase amount by selected shipping type.

AVERAGE PURCHASE BY SHIPPING TYPE



Observed order value; this is an association, not an effect of shipping choice.

AVERAGE DIFFERENCE

+$2.02

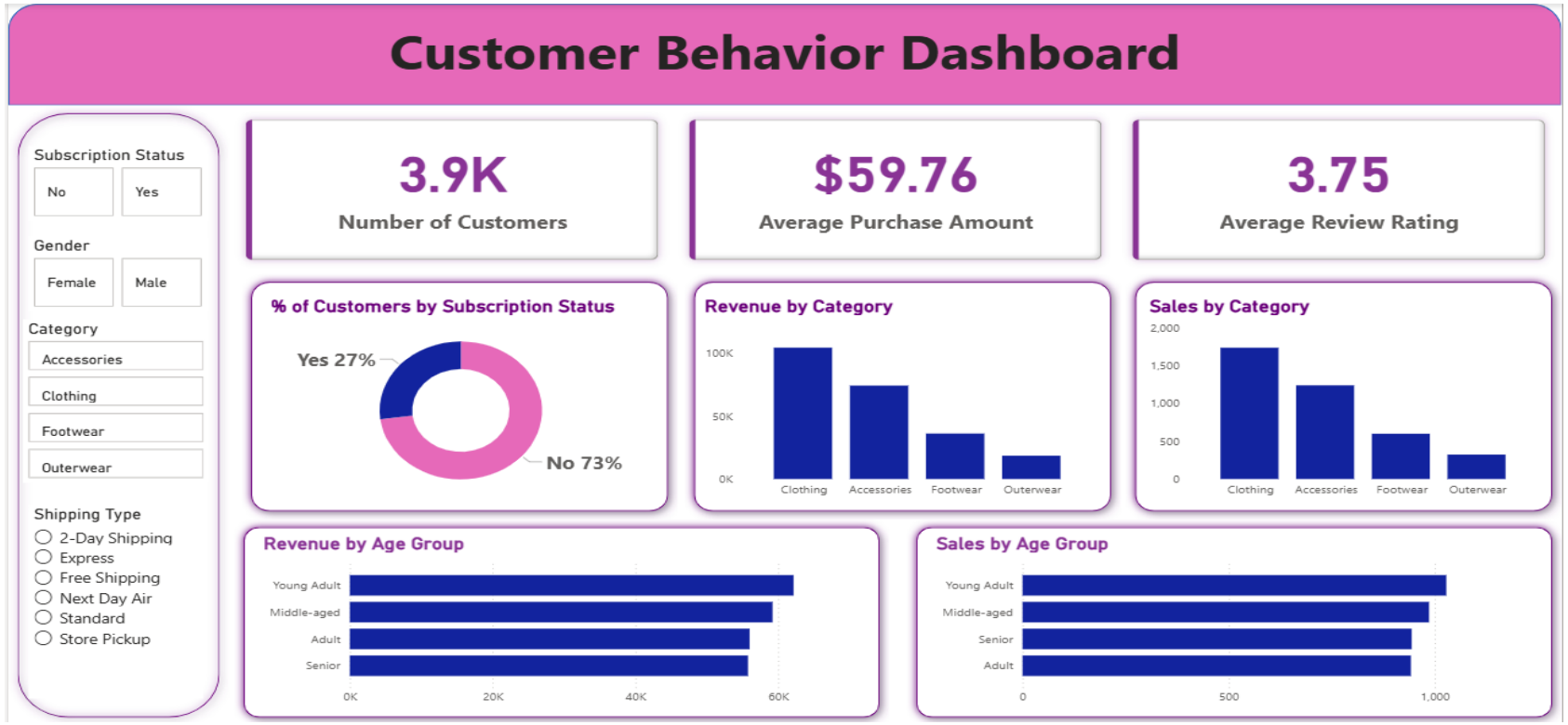
DISCOUNT BEHAVIOR

839 discounted records above the overall average purchase

Source: project report

The analysis comes together in Power BI

The source project presents an interactive dashboard for filtering and communicating the findings.



Dashboard screenshot reproduced from the original project report.

DASHBOARD KPIS

3.9K
Customers

$59.76
Avg. purchase

3.75
Avg. review

Turn descriptive patterns into testable actions

Five targeted initiatives connect the findings to measurable commercial outcomes.

01	**Grow subscription adoption**	73% of records are non-subscribed.	**Pilot a compelling sign-up benefit and track conversion.**
02	**Develop loyalty journeys**	3,116 records fall into the Loyal segment.	**Reward repeat behavior and monitor return frequency.**
03	**Feature strong products**	Gloves rate 3.86; Jewelry and Blouse rank #1.	**Test top-rated and popular products in campaigns.**
04	**Audit discount economics**	Hat purchases use discounts 50% of the time.	**Compare promotion lift with contribution margins.**
05	**Test targeted creatives**	Young adults contribute $62.1K in revenue.	**Compare age-targeted messaging and conversion.**

FROM INSIGHT TO IMPACT.

The next step is to validate which actions improve conversion, retention, and margin.

01 Measurement

37 missing review ratings were imputed by category median.

02 Interpretation

Comparisons are descriptive; they do not establish causality.

03 Next data

Add profitability, time-series behavior, and experiment outcomes.

PYTHON → POSTGRESQL → POWER BI